JACOBIAN LOCALIZATION AND REGULAR SEQUENCES IN TWO VARIABLES

Abstract. For polynomials F, G over any commutative ring, their Jacobian determinant J has the following property: $FU = GV$ implies $J^e U = GW$ for some $e \geq 0$ and some polynomial W. We prove this by a finite Taylor lifting construction and stabilization of annihilators, after reduction to a noetherian coefficient ring. Together with a content argument, this proves that (F, G) is regular in both orders after inverting J. The proof uses no flatness theorem. The final section describes the Unit formalisation, which proves the divisibility assertion under an explicit annihilator-stabilization hypothesis; the noetherian reduction remains outside the formalisation.

1. Statement

All rings are commutative with identity; the zero ring is allowed. Let A be a ring, let $R = A[X, Y]$, and let $F, G \in R$. Write

$$J = F_X G_Y - F_Y G_X, \qquad S = R[1/J],$$

where subscripts denote formal partial derivatives. An element is *regular* if multiplication by it is injective. We call (a, b) *commutatively regular* if a, b are regular individually, b is regular modulo (a), and a is regular modulo (b). Here regularity of a sequence does not require its generated ideal to be proper.

Theorem 1.1. *The pair (F, G) is commutatively regular in S. Moreover, for all $U, V \in R$,*

$$FU = GV \quad \Longrightarrow \quad \exists e \geq 0 \; \exists W \in R \quad J^e U = GW. \tag{1}$$

The individual regularity of F and G is Step I. The principal point is Step II, namely (1). We first give a conditional proof of Step II using only finite polynomial calculations, then supply its hypothesis by a noetherian reduction.

2. Step I: content and individual regularity

For a polynomial H in finitely many indeterminates over A, let $\mathrm{c}(H) \subseteq A$ denote the ideal generated by its coefficients. The following argument is the one in the supplied note `CAR1.pdf` [1].

Lemma 2.1. *Let $H, P \in A[\mathbf{X}]$. If $\mathrm{c}(P) \subseteq \mathrm{c}(H)$, then H is regular in $A[\mathbf{X}][1/P]$.*

Proof. We use the consequence of Dedekind–Mertens that, for polynomials H, Q, there is an integer m such that

$$\mathrm{c}(H)^m \mathrm{c}(Q) \subseteq \mathrm{c}(HQ).$$

Suppose a polynomial K satisfies $HK = 0$ after localization. Then $P^k HK = 0$ in $A[\mathbf{X}]$ for some $k \geq 0$. Apply the displayed inclusion to $Q = P^k K$. For a suitable m,

$$\mathrm{c}(P^{m+k}K) \subseteq \mathrm{c}(P)^m \mathrm{c}(P^k K) \subseteq \mathrm{c}(H)^m \mathrm{c}(P^k K) \subseteq \mathrm{c}(HP^k K) = 0.$$

Hence $P^{m+k}K = 0$, so $K = 0$ after localization. Every localized element has a polynomial numerator, which proves the assertion. □

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Every coefficient of a partial derivative of F belongs to $\mathrm{c}(F)$, and similarly for G. Consequently

$$\mathrm{c}(J) \subseteq \mathrm{c}(F), \qquad \mathrm{c}(J) \subseteq \mathrm{c}(G).$$

Lemma 2.1, with $P = J$, proves the individual regularity of F and G in S. Notice the direction of the content inclusion in the lemma.

3. Two finite algebraic lemmas

Definition 3.1. For a ring B, an element $f \in B$ and an integer $N \geq 0$, write $\mathrm{Stable}_B(f, N)$ for the condition

$$\forall v \in B, \qquad f^{N+1}v = 0 \implies f^N v = 0. \tag{2}$$

Equivalently, $\mathrm{Ann}_B(f^N) = \mathrm{Ann}_B(f^{N+1})$.

Lemma 3.2 (A truncated linear equation)**.** *Assume* $\mathrm{Stable}_B(f, N)$*. If*

$$(f - T)p(T) = 0 \pmod{T^{N+1}},$$

then $p(0) = 0$*.*

Proof. Write $p(T) = u_0 + u_1 T + \cdots + u_N T^N$ modulo T^{N+1}. Comparison of coefficients gives

$$fu_0 = 0, \qquad fu_i = u_{i-1} \quad (1 \leq i \leq N).$$

Thus $u_0 = f^N u_N$ and $f^{N+1}u_N = 0$. Condition (2) implies $f^N u_N = 0$, hence $u_0 = 0$. This also covers $N = 0$: the hypothesis then says that f itself is regular. □

Lemma 3.3 (Finite Taylor lifting)**.** *Let* B *be an* A*-algebra, and let* $x, y \in B$ *satisfy*

$$G(x, y) = 0, \qquad J(x, y) \in B^\times.$$

Put $f = F(x, y)$*. For every* $N \geq 0$*, there are polynomials* $x_N(T), y_N(T) \in B[T]$ *such that*

$$x_N(0) = x, \qquad y_N(0) = y, \tag{3}$$

$$F(x_N(T), y_N(T)) \equiv f - T \pmod{T^{N+1}}, \qquad G(x_N(T), y_N(T)) \equiv 0 \pmod{T^{N+1}}. \tag{4}$$

Proof. Start with the constant polynomials x, y, which give the assertions modulo T. Suppose x_*, y_* have already been constructed modulo T^r, where $r \geq 1$, and write their next errors as

$$F(x_*, y_*) - (f - T) \equiv \epsilon_F T^r, \qquad G(x_*, y_*) \equiv \epsilon_G T^r \pmod{T^{r+1}}.$$

We seek corrections $x_* + aT^r$ and $y_* + bT^r$. For every polynomial H, the ordinary polynomial expansion gives

$$H(x_* + aT^r, y_* + bT^r) \equiv H(x_*, y_*) + T^r\big(H_X(x, y)a + H_Y(x, y)b\big) \pmod{T^{r+1}}. \tag{5}$$

Indeed, terms containing at least two corrections are divisible by T^{2r}, hence by T^{r+1}; in the linear terms, the derivatives may be evaluated at x, y since x_*, y_* have those constant terms. This argument is valid in every characteristic.

The matrix

$$M = \begin{pmatrix} F_X(x, y) & F_Y(x, y) \\ G_X(x, y) & G_Y(x, y) \end{pmatrix}$$

has determinant $J(x, y)$, a unit. Therefore we can take

$$\begin{pmatrix} a \\ b \end{pmatrix} = -M^{-1}\begin{pmatrix} \epsilon_F \\ \epsilon_G \end{pmatrix}, \qquad M^{-1} = J(x, y)^{-1}\begin{pmatrix} G_Y(x, y) & -F_Y(x, y) \\ -G_X(x, y) & F_X(x, y) \end{pmatrix}.$$

By (5), both errors then vanish modulo T^{r+1}. The constant terms are unchanged. Performing these corrections for $r = 1, \ldots, N$ proves the lemma. For $N = 0$ the initial constants suffice. □

The construction uses only finitely many polynomial operations and the inverse of $J(x, y)$. In particular, it does not require division by factorials or the construction of an infinite formal power series.

4. Step II under the stabilization hypothesis

Let

$$B = \big(R/(G)\big)[1/J],$$

where the localization is at the image of J. Denote the images of X, Y, F, U in B by x, y, f, u, respectively.

Proposition 4.1. *Suppose there exists $N \geq 0$ with $\mathrm{Stable}_B(f, N)$. Then (1) holds for every $U, V \in R$.*

Proof. Assume $FU = GV$. In B we have $G(x, y) = 0$ and $J(x, y)$ is a unit, so Lemma 3.3 supplies x_N, y_N. Substitution into the polynomial identity $FU = GV$ gives

$$(f - T)U(x_N(T), y_N(T)) \equiv 0 \pmod{T^{N+1}}.$$

Lemma 3.2 implies

$$u = U(x_N(0), y_N(0)) = 0 \quad \text{in } B.$$

The kernel criterion for localization now gives $e \geq 0$ such that $J^e U = 0$ in $R/(G)$. Thus $J^e U \in (G)$ in R, which is precisely $J^e U = GW$ for some $W \in R$. □

5. Noetherian reduction and completion of the proof

Proof of (1) over an arbitrary ring. Fix a relation $FU = GV$. Let $A_0 \subseteq A$ be the subring generated by the coefficients of these four polynomials and by 1. It is a finitely generated $\mathbb{Z}$-algebra, hence noetherian by the Hilbert basis theorem. Since A_0 is a subring, the given identity already holds in $A_0[X, Y]$.

Set

$$B_0 = \big(A_0[X, Y]/(G)\big)[1/J], \qquad f_0 = \text{the image of } F \text{ in } B_0.$$

Polynomial extensions, quotients and localizations of noetherian rings are noetherian. Hence the ascending chain

$$\mathrm{Ann}_{B_0}(1) \subseteq \mathrm{Ann}_{B_0}(f_0) \subseteq \mathrm{Ann}_{B_0}(f_0^2) \subseteq \cdots$$

stabilizes. In particular, $\mathrm{Stable}_{B_0}(f_0, N)$ holds for some N. Proposition 4.1, applied over A_0, gives $e \geq 0$ and $W \in A_0[X, Y]$ with $J^e U = GW$. This identity also holds in $A[X, Y]$. □

This reduction supplies the stabilization hypothesis in B_0 for the particular relation under consideration. It does not assert that the ring B formed over the original arbitrary A is noetherian. The finite argument of Section 4 is constructive under its stated hypothesis; the reduction just given uses classical noetherian algebra.

Completion of Theorem 1.1. Step I gives the individual regularity of F and G in S. To see that F is regular modulo G, work in $S/(G) \cong (R/(G))[1/J]$. Suppose an element represented by U/J^r is killed by F. Clearing the localization equality gives

$$J^k FU = GV$$

in R for some $k \geq 0$ and $V \in R$. Apply (1) to $F(J^k U) = GV$. We obtain $J^{e+k} U \in (G)$ for some e, so $U/J^r = 0$ in $S/(G)$.

Exchanging F and G changes the Jacobian to $-J$, which defines the same localization. The same argument therefore proves that G is regular modulo F. This proves regularity in both orders. □

6. What has been formalised in Unit

The development is in `unit/sag/jacobian/`. The theorem checked by Unit is Proposition 4.1, including the construction of B and the final certificate in $A[X,Y]$. Its only additional algebraic hypothesis is

$$\left\| \sum_{N:\mathbb{N}} \mathrm{Stable}_B(f,N) \right\|. \tag{6}$$

Here $\|\cdot\|$ denotes propositional truncation, or mere existence. The public theorem takes an actual polynomial identity $FU = GV$ and returns

$$\left\| \sum_{e:\mathbb{N}} \sum_{W:A[X,Y]} (J^e U = GW) \right\|. \tag{7}$$

Thus neither N nor the output witnesses need be chosen globally. The untruncated type inside (6) is named `NoetherianInput`; its name does not assert a formalised noetherianity theorem.

Polynomial representation and the public interface. The source ring is the existing coefficient-quotient polynomial ring $A[X][Y]$. The inputs F, G are finite polynomial expressions with constants in A, two variables, sums and products. Partial derivatives are evaluated by the usual formal sum and product rules. `ExpressionRepresentation` proves that every element of $A[X][Y]$ merely has such an expression. The inputs U, V to the public theorem are arbitrary elements of this polynomial ring. All identity hypotheses are equalities in that ring, rather than equalities of polynomial functions.

The public interface, with the foundation parameters understood, is

```
module D = JacobianDivisibility[l].Over[A,setA,F,G]
D.result input u v relation
```

Here `setA` expresses that the carrier of the commutative ring is a set, `input` has type (6), and `relation` is the equality $FU = GV$.

Correspondence with the proof. The following table lists the source files in the development directory. Each file name has extension `.seg`.

File or files	Checked argument
`FiniteJets`	Equality of coefficients below a finite degree, and the product rule for changing a coefficient of positive degree.
`StableAnnihilator`	Lemma 3.2, including $N = 0$.
`Bivariate`, `BivariateNaturality`	Polynomial expressions, evaluation and evaluated partial derivatives; Taylor's correction formula and compatibility with ring maps.
`LinearSolve`, `TaylorLift`	Inversion of the 2×2 Jacobian matrix and the finite construction of Lemma 3.3.
`PolynomialIdentity`, `RegularAt`	Substitution of the genuine polynomial identity into the finite lift, and the conclusion $u = 0$ in the target ring.
`PrincipalQuotient`, `PowerLocalization`	The quotient by G and localization at J, with their proved kernel certificates. The localization reuses the existing fraction library.

File or files	Checked argument
`ExpressionRepresentation`	Expression representatives for all elements of $A[X][Y]$, by quotient and finite list induction.
`JacobianDivisibility`	Proposition 4.1 and the conclusion (7), after clearing denominators.
`Examples`	For $F = X + X^2$, $G = Y$ at $(0, 0)$, all finite lifts and the coefficients $x(T) = -T - T^2 \bmod T^3$, $y(T) = 0 \bmod T^3$, over any commutative ring.

Foundations, validation and remaining work. The construction uses the same univalence and replacement parameters as the existing Unit polynomial library, from which the effective set quotients and propositional truncations are obtained. It introduces no excluded-middle or choice axiom. All recursive definitions use Unit's data eliminators. `JetIdentities.seg` contains ordinary equality proofs generated by the existing ring-identity generator. Unit checks these proof terms; the generator is not an additional trusted principle.

From `unit/`, the complete checking command is

```
./segc sag/jacobian all
```

The check on October 25, 2026 reported *40 modules, 0 failed*. This count includes the imported library modules.

The remaining unformalised part is the derivation of (6) by the noetherian argument in Section 5: the finite coefficient reduction, the Hilbert basis theorem and preservation of noetherianity, and stabilization of the annihilator chain. Step I from Section 2 is also not formalised in this directory. The separate development `unit/sag/mertens/` now proves the univariate Dedekind–Mertens formula over every commutative ring, following the supplied direct proof [2]. It constructs signed-sum certificates for the stronger additive-subgroup inclusion, proves the reverse inclusion, and obtains both ideal inclusions. Its check reports *14 modules, 0 failed*. The multivariate content lemma and localization application used in Step I have not yet been connected to that result. A separate note, `dedekind-mertens.tex`, presents the proof and its formalisation status.

Consequently the unrestricted Theorem 1.1 is proved in this note, while the checked Unit result at present is the conditional Step II described above.

References

[1] *ContentAndRegularity.tex*, one-page note supplied as `CAR1.pdf`, compilation dated October 25, 2026.
[2] *A direct proof of the Dedekind-Mertens Lemma*, supplied as `mertens.pdf`, April 5, 2006.